221-2327 Quiz M.Rohaan Sec-E

Q#01:-

a) $k+2k+3k+4k=1$

$10k=1$

$\boxed{k=1/10}$

b) $P(X\leq 2)=P(0)+P(1)+P(2)=k+2k+3k=6k$

$=\frac{6}{10}=\frac{3}{5}$

c) $E(X)=\sum x\cdot P(X=x)=0\cdot k+1\cdot 2\cdot\frac{1}{10}+2\cdot 3\cdot\frac{1}{10}+$

$3\cdot 4\cdot\frac{1}{10}=2$

Q#02:-

a) $P(X=x)=\begin{cases}\frac{x}{10}, & x=1,2,3,4\\ 0, & \text{otherwise}\end{cases}$

b) $P(X>2)=P(3)+P(4)=\frac{3}{10}+\frac{4}{10}=\frac{7}{10}$

c) $V(X)=E(X^2)-(E(X))^2$

$$E(X) = 1\cdot\frac{1}{10} + 2\cdot\frac{2}{10} + 3\cdot\frac{3}{10} + 4\cdot\frac{4}{10} = 3$$

$$E(X^2) = 1^2\cdot\frac{1}{10} + 2^2\cdot\frac{2}{10} + 3^2\cdot\frac{3}{10} + 4^2\cdot\frac{4}{10} = 10$$

$$V(X) = 10 - 3^2 = 1$$

Q#03:

a) $$\int_0^1 c(1-x)\,dx = 1$$

$$c\int_0^1 (1-x)\,dx = 1$$

$$c\left[x - \frac{x^2}{2}\right]_0^1 = 1 \Rightarrow c\left(1 - \frac{1}{2}\right) = 1$$

$$c = 2$$

b) $$P\left(\frac{1}{2} \le X \le 1\right) = \int_{1/2}^1 2(1-x)\,dx$$

$$2\left[x - \frac{x^2}{2}\right]_{1/2}^1 = 2\left[1 - \frac{1^2}{2} - \left(\frac{1}{2} - \frac{\left(\frac{1}{2}\right)^2}{2}\right)\right]$$

$$= \frac{1}{4}$$

c) $$E(X) = \int_0^1 x\cdot 2(1-x)\,dx$$

$$= 2\int_0^1 (x - x^2)\,dx = 2\left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1$$

$$= 2\left[\frac{1}{2} - \frac{1}{3}\right] = \frac{1}{3}$$

Q#04:-

a) $f(x) = \frac{d}{dx}F(x) \Rightarrow f(x) = \frac{d}{dx}(x^2) = 2x$

$$f(x) = \begin{cases} 2x, & 0 \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$$

b) $P\left(\frac{1}{2} \le X \le 1\right) = F(1) - F\left(\frac{1}{2}\right) = 1^2 - \left(\frac{1}{2}\right)^2 = 1 - \frac{1}{4} = \frac{3}{4}$

c) $F(m) = \frac{1}{2} \Rightarrow m^2 = \frac{1}{2} \Rightarrow \boxed{m = \frac{1}{\sqrt{2}}}$

Q#05:-

a) $\int_0^3 kx\,dx = 1$

$$k\left[\frac{x^2}{2}\right]_0^3 = 1 \Rightarrow k\left[\frac{3^2}{2}\right] \Rightarrow k = \frac{2}{9}$$

b) $P(X > 1) = \int_1^3 kx\,dx = \int_1^3 \frac{2}{9}x\,dx$

$$\frac{2}{9}\left[\frac{x^2}{2}\right]_1^3 = \frac{2}{9}\left[\frac{3^2}{2} - \frac{1^2}{2}\right] = \frac{8}{9}$$

c) $E(X^2) = \int_0^3 x^2 \cdot kx \, dx = \frac{2}{9}\int_0^3 x^3 \, dx$

$= \frac{2}{9}\left[\frac{x^4}{4}\right]_0^3 = \frac{2}{9}\left[\frac{3^4}{4}\right] = \frac{9}{2}$